Stochastic Processes - Homework 1

Pranav Tikkawar

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Problem 1: σ-algebras

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ with the uniform probability measure (a fair die). Define

$$Z(\omega) = \begin{cases} 0, & \omega \in \{1, 2, 3\}, \\ 1, & \omega \in \{4, 5, 6\}. \end{cases}$$

Let $\mathcal{G} = \sigma(Z)$.

1(a) List all sets in $\mathcal{G}$ and identify its atoms

$\mathcal{G} = \{\emptyset, \{1, 2, 3\}, \{4, 5, 6\}, \Omega\}$

The atoms of $\mathcal{G}$ are $\{1, 2, 3\}$ and $\{4, 5, 6\}$.

1(b) Measurability of $X(\omega) = \mathbf{1}\{\omega \text{ is even}\}$

To test if X is $\mathcal{G}$-measurable, we check if the preimage of any set generated by the values of X is in $\mathcal{G}$. The function X takes the value 1 for even ω (2, 4, 6) and 0 for odd ω (1, 3, 5).

The preimage of $\{1\}$ under X is $\{2, 4, 6\}$, which is not in $\mathcal{G}$. Therefore, X is not $\mathcal{G}$-measurable.

1(c) Compute $\mathbb{E}[X \mid \mathcal{G}]$

To compute $\mathbb{E}[X \mid \mathcal{G}]$, we need to find the conditional expectation of X given the σ-algebra $\mathcal{G}$. Since $\mathcal{G}$ has two atoms, we compute the expectation on each atom.

$$E[X \mid \mathcal{G}](\omega) = \begin{cases} E[X \mid Z = 0], & \omega \in \{1, 2, 3\}, \\ E[X \mid Z = 1], & \omega \in \{4, 5, 6\}. \end{cases} E[X \mid Z = 0] \quad = \frac{1}{3}(X(1) + X(2) + X(3)) = \frac{1}{3}(0 + 1 + 0)$$

$E[X \mid Z = 1] = \frac{1}{3}(X(4) + X(5) + X(6)) = \frac{1}{3}(1 + 0 + 1) = \frac{2}{3}.$

Thus,

$$\mathbb{E}[X \mid \mathcal{G}](\omega) = \begin{cases} \frac{1}{3}, & \omega \in \{1, 2, 3\}, \\ \frac{2}{3}, & \omega \in \{4, 5, 6\}. \end{cases}$$

1(d) Compute $\mathbb{E}[Y \mid \mathcal{G}]$ for $Y(\omega) = \omega$

To compute $\mathbb{E}[Y \mid \mathcal{G}]$, we again compute the expectation on each atom of $\mathcal{G}$.

$$E[Y \mid Z = 0] = \frac{1}{3}(Y(1) + Y(2) + Y(3)) = \frac{1}{3}(1 + 2 + 3) = 2,$$
$$E[Y \mid Z = 1] = \frac{1}{3}(Y(4) + Y(5) + Y(6)) = \frac{1}{3}(4 + 5 + 6) = 5.$$

Thus,

$$\mathbb{E}[Y \mid \mathcal{G}](\omega) = \begin{cases} 2, & \omega \in \{1, 2, 3\}, \\ 5, & \omega \in \{4, 5, 6\}. \end{cases}$$

Problem 2: A martingale from prediction (Doob martingale)

Let $\xi_1, \ldots, \xi_6$ be i.i.d. Bernoulli(1/2) (1 = head, 0 = tail). Let

$$\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n), \qquad S_n = \sum_{i=1}^{n} \xi_i.$$

Define

$$X = \mathbf{1}\Big\{ \sum_{i=1}^{6} \xi_i \geq 4 \Big\}, \qquad M_n = \mathbb{E}[X \mid \mathcal{F}_n], \quad n = 0, 1, \ldots, 6.$$

2(a) Closed-form expression for M_n in terms of S_n

This is the conditional expectaion of X given the first n coin flips. Given $\mathcal{F}_n$ we know S_n, the number of heads in the first n flips.

Define

$$K := \sum_{i=n+1}^{6} \xi_i, \sim \text{Binomial}(6 - n, 1/2)$$

Thus $\sum_{i=1}^{6} \xi_i = S_n + K$.

So then we have

$$M_n = \mathbb{P}(S_n + K \geq 4 | \mathcal{F}_n) = \mathbb{P}(K \geq 4 - S_n)$$
$$= \sum_{k=\max(0,4-S_n)}^{6-n} \binom{6-n}{k} \left(\frac{1}{2}\right)^{6-n}.$$

2(b) Compute M_0

We want $M_0 = \mathbb{E}[X \mid \mathcal{F}_0] = \mathbb{E}[X]$. Since X is the indicator of the event that at least 4 heads occur in 6 flips, we can compute this probability directly:

$$\begin{aligned}
M_0 = \mathbb{P}(S_6 \geq 4) &= \sum_{k=4}^{6} \binom{6}{k} \left(\frac{1}{2}\right)^6 \\
&= \binom{6}{4} \left(\frac{1}{2}\right)^6 + \binom{6}{5} \left(\frac{1}{2}\right)^6 + \binom{6}{6} \left(\frac{1}{2}\right)^6 \\
&= \left(\frac{1}{2}\right)^6 \left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right) \\
&= \left(\frac{1}{2}\right)^6 (15 + 6 + 1) = \frac{22}{64} = \frac{11}{32}.
\end{aligned}$$

2(c) Compute M_3 on the event $\{\xi_1 = 1, \xi_2 = 0, \xi_3 = 1\}$

On the event $\{\xi_1 = 1, \xi_2 = 0, \xi_3 = 1\}$, we have $S_3 = 2$. Thus, $S_3 = 2$ and we need to compute

$$\begin{aligned}
M_3 &= \mathbb{P}(S_3 + K \geq 4 | \mathcal{F}_3) = \mathbb{P}(K \geq 4 - S_3) = \mathbb{P}(K \geq 2) \\
&= \sum_{k=2}^{3} \binom{3}{k} \left(\frac{1}{2}\right)^3 = \binom{3}{2} \left(\frac{1}{2}\right)^3 + \binom{3}{3} \left(\frac{1}{2}\right)^3 \\
&= \left(\frac{1}{2}\right)^3 (3 + 1) = \frac{4}{8} = \frac{1}{2}.
\end{aligned}$$

2(d) Prove (M_n) is a martingale w.r.t. $(\mathcal{F}_n)$

To show that (M_n) is a martingale with respect to $(\mathcal{F}_n)$, we need to verify the following conditions:

- $\mathbb{E}[|M_n|] < \infty$ for all n.
- M_n is $\mathcal{F}_n$-measurable for all n.
- $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ for all n.

1. Since M_n is a conditional expectation of an indicator random variable, it takes values in $[0, 1]$. Thus, $\mathbb{E}[|M_n|] \leq 1 < \infty$.

2. By definition, $M_n = \mathbb{E}[X \mid \mathcal{F}_n]$ is $\mathcal{F}_n$-measurable.

3. We need to show that $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$. Using the tower property of conditional expectation, we have

$$\begin{aligned}
\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\
&= \mathbb{E}[X \mid \mathcal{F}_n] = M_n.
\end{aligned}$$

Thus, (M_n) is a martingale with respect to $(\mathcal{F}_n)$.

Problem 3: Wald's lemma and the negative binomial mean

Let $\xi_1, \xi_2, \ldots$ be i.i.d. Bernoulli(p) with $p \in (0, 1)$. Let

$$S_n = \sum_{i=1}^{n} \xi_i, \qquad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n).$$

Fix an integer $r \geq 1$ and define the stopping time

$$T = \inf\{n \geq 0 : S_n = r\}.$$

3(a) Show that T is a stopping time w.r.t. $(\mathcal{F}_n)$

A process T is a stopping time with respect to the filtration $(\mathcal{F}_n)$ if for every $n \geq 0$, the event $\{T = n\}$ is in $\mathcal{F}_n$.

Fix $n \geq 0$. The event $\{T = n\}$ can be expressed as

$$\{T = n\} = \{S_n = r\} \cap \{S_k < r \text{ for all } k < n\}.$$

Each S_k is a function of $\xi_1, \ldots, \xi_k$, and thus is $\mathcal{F}_k$-measurable. Since $\mathcal{F}_k \subseteq \mathcal{F}_n$ for $k < n$, the event $\{S_k < r\}$ is also in $\mathcal{F}_n$. Therefore, the intersection of these events is in $\mathcal{F}_n$, which means $\{T = n\} \in \mathcal{F}_n$. Since this holds for all n, T is a stopping time with respect to $(\mathcal{F}_n)$.

3(b) Show that $Z_n = S_n - pn$ is a martingale w.r.t. $(\mathcal{F}_n)$

To show that $Z_n = S_n - pn$ is a martingale with respect to $(\mathcal{F}_n)$, we need to verify the following conditions:

- $\mathbb{E}[|Z_n|] < \infty$ for all n.
- Z_n is $\mathcal{F}_n$-measurable for all n.
- $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n$ for all n.

1. Since S_n is a sum of n Bernoulli random variables, it takes values in $\{0, 1, \ldots, n\}$. Thus, $|Z_n| = |S_n - pn| \leq n + pn < \infty$, so $\mathbb{E}[|Z_n|] < \infty$. 2. Z_n is a function of $\xi_1, \ldots, \xi_n$, so it is $\mathcal{F}_n$-measurable. 3. We need to show that $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n$. We have

$$\begin{aligned}
\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_{n+1} - p(n+1) \mid \mathcal{F}_n] \\
&= \mathbb{E}[S_n + \xi_{n+1} - p(n+1) \mid \mathcal{F}_n] \\
&= S_n - pn + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] - p \\
&= S_n - pn + p - p = Z_n.
\end{aligned}$$

Thus, Z_n is a martingale with respect to $(\mathcal{F}_n)$.

3(c) Apply bounded Wald's lemma to Z_{T_N}

Define $T_N = T \wedge N$. Since T_N is a bounded stopping time (it is at most N), we can apply the bounded version of Wald's lemma to the martingale (Z_n) at the stopping time T_N.

By Wald's lemma, we have

$$\mathbb{E}[Z_{T_N}] = \mathbb{E}[Z_0]$$

Since $Z_0 = S_0 - p \cdot 0 = 0$ and $Z_{T_N} = S_{T_N} - pT_N$, we get

$$\begin{aligned} &\mathbb{E}[Z_{T_N}] = \mathbb{E}[S_{T_N} - pT_N] = E[S_{T_N}] - p\mathbb{E}[T_N] = 0 \\ \Rightarrow\ &\mathbb{E}[S_{T_N}] = p\mathbb{E}[T_N]. \end{aligned}$$

Thus we have shown that $\mathbb{E}[S_{T_N}] = p\mathbb{E}[T_N]$.

3(d) Take the limit $N \to \infty$ and conclude $\mathbb{E}[T] = r/p$

As $N \to \infty$, the stopping time $T_N = T \wedge N$ converges to T almost surely. Since $S_{T_N} = r$ for all N large enough (specifically, for all $N \geq T$), we have

$$\mathbb{E}[S_{T_N}] = r.$$

Taking the limit as $N \to \infty$ in the equation $\mathbb{E}[S_{T_N}] = p\mathbb{E}[T_N]$, we get

$$r = p \lim_{N\to\infty} \mathbb{E}[T_N] = p\mathbb{E}[T].$$

Thus, we conclude that

$$\mathbb{E}[T] = \frac{r}{p}.$$